

Representative Structural Calculation Package

L-shaped neighborhood facility

Item	Value
Project	L-shaped neighborhood facility
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	02-l-shaped-neighborhood

Nodes	Members	Loads	Combos	Max disp.	Max util.
48	84	240	28	3.715 mm	0.363

Model And Design Basis

Item	Value
Building type	neighborhood-living
Description	Three-story L-shaped low-rise commercial frame with plan irregularity.
Occupancy	Office
Bounds X/Y/Z	16.50 / 13.50 / 10.50
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	48	0.000 kN	0.000 kN	-2673 kN
L	live	48	0.000 kN	0.000 kN	-1336 kN
WX	wind	36	63.79 kN	0.000 kN	0.000 kN
WY	wind	36	0.000 kN	77.96 kN	0.000 kN
EX	seismic	36	219.52 kN	0.000 kN	0.000 kN
EY	seismic	36	0.000 kN	219.52 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	222.75	891.00 kN	534.60 kN	21.26 kN	25.99 kN
2	222.75	891.00 kN	534.60 kN	21.26 kN	25.99 kN
3	222.75	891.00 kN	267.30 kN	21.26 kN	25.99 kN

Load Derivation Formula Trace

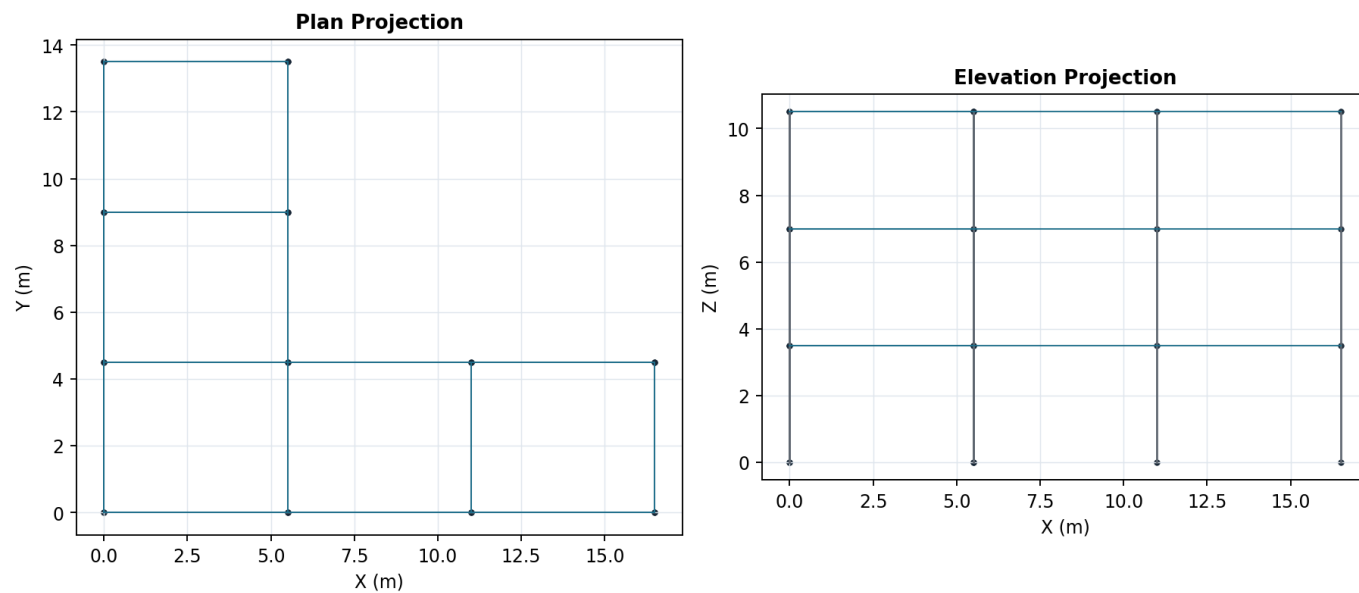
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=office
basis-plan-area	-	$\max(\text{model width} * \text{model depth}, 1)$	222.75 m2	Bx=16.50 m, By=13.50 m
basis-roof-area	-	roofArea input or typical floor area	222.75 m2	roofArea=222.75 m2
D-ST1	D	$A * q_D$	891.00 kN	$A=222.75 \text{ m}^2$, $q_D=4.000 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	534.60 kN	$A=222.75 \text{ m}^2$, $q_L=2.400 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	21.26 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=13.50 \text{ m}$, $h=3.500 \text{ m}$
WY-ST1	WY	$p_{WY} * B_x * h$	25.99 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=16.50 \text{ m}$, $h=3.500 \text{ m}$
D-ST2	D	$A * q_D$	891.00 kN	$A=222.75 \text{ m}^2$, $q_D=4.000 \text{ kN/m}^2$
L-ST2	L	$A * q_L$	534.60 kN	$A=222.75 \text{ m}^2$, $q_L=2.400 \text{ kN/m}^2$
WX-ST2	WX	$p_{WX} * B_y * h$	21.26 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=13.50 \text{ m}$, $h=3.500 \text{ m}$
WY-ST2	WY	$p_{WY} * B_x * h$	25.99 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=16.50 \text{ m}$, $h=3.500 \text{ m}$
D-ST3	D	$A * q_D$	891.00 kN	$A=222.75 \text{ m}^2$, $q_D=4.000 \text{ kN/m}^2$
L-ST3	L	$A * q_L$	267.30 kN	$A=222.75 \text{ m}^2$, $q_L=1.200 \text{ kN/m}^2$
WX-ST3	WX	$p_{WX} * B_y * h$	21.26 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=13.50 \text{ m}$, $h=3.500 \text{ m}$

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

L-shaped neighborhood facility



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	1.029 mm	0.128	0.000
KDS-ST-02	OK	1.299 mm	0.182	0.000
KDS-ST-04-WX-P	OK	1.414 mm	0.179	0.000
KDS-ST-04-WX-N	OK	1.409 mm	0.175	0.000
KDS-ST-04-WY-P	OK	1.391 mm	0.185	0.000
KDS-ST-04-WY-N	OK	1.446 mm	0.182	0.000
KDS-ST-05-EX-P	OK	3.715 mm	0.248	0.000
KDS-ST-05-EX-N	OK	3.683 mm	0.243	0.000
KDS-ST-05-EY-P	OK	3.385 mm	0.246	0.000
KDS-ST-05-EY-N	OK	3.391 mm	0.241	0.000
KDS-ST-06-WX-P	OK	1.063 mm	0.108	0.000
KDS-ST-06-WX-N	OK	1.059 mm	0.105	0.000
KDS-ST-06-WY-P	OK	1.111 mm	0.113	0.000
KDS-ST-06-WY-N	OK	1.118 mm	0.111	0.000
KDS-ST-07-EX-P	OK	3.623 mm	0.176	0.000
KDS-ST-07-EX-N	OK	3.606 mm	0.173	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	OK
Max drift	1.489 mm
Max drift ratio	0.00043
Governing	KDS-ST-05-EX-P / story 2

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	3.500 m	0.013 mm	0.00000	0.001	OK
KDS-ST-01	2	3.500 m	0.018 mm	0.00001	0.001	OK
KDS-ST-01	3	3.500 m	0.047 mm	0.00001	0.003	OK
KDS-ST-02	1	3.500 m	0.020 mm	0.00001	0.001	OK
KDS-ST-02	2	3.500 m	0.026 mm	0.00001	0.002	OK
KDS-ST-02	3	3.500 m	0.059 mm	0.00002	0.003	OK
KDS-ST-04-WX-P	1	3.500 m	0.300 mm	0.00009	0.017	OK
KDS-ST-04-WX-P	2	3.500 m	0.365 mm	0.00010	0.021	OK
KDS-ST-04-WX-P	3	3.500 m	0.240 mm	0.00007	0.014	OK
KDS-ST-04-WX-N	1	3.500 m	0.296 mm	0.00008	0.017	OK
KDS-ST-04-WX-N	2	3.500 m	0.354 mm	0.00010	0.020	OK
KDS-ST-04-WX-N	3	3.500 m	0.234 mm	0.00007	0.013	OK
KDS-ST-04-WY-P	1	3.500 m	0.343 mm	0.00010	0.020	OK
KDS-ST-04-WY-P	2	3.500 m	0.404 mm	0.00012	0.023	OK
KDS-ST-04-WY-P	3	3.500 m	0.256 mm	0.00007	0.015	OK
KDS-ST-04-WY-N	1	3.500 m	0.347 mm	0.00010	0.020	OK
KDS-ST-04-WY-N	2	3.500 m	0.405 mm	0.00012	0.023	OK
KDS-ST-04-WY-N	3	3.500 m	0.255 mm	0.00007	0.015	OK
KDS-ST-05-EX-P	1	3.500 m	1.077 mm	0.00031	0.062	OK
KDS-ST-05-EX-P	2	3.500 m	1.489 mm	0.00043	0.085	OK

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M6	OK	0.363	steel-interaction	KDS-ST-05-EX-P
2	M7	OK	0.329	steel-interaction	KDS-ST-05-EX-N
3	M3	OK	0.328	steel-interaction	KDS-ST-05-EX-N
4	M10	OK	0.326	steel-interaction	KDS-ST-05-EY-N
5	M9	OK	0.323	steel-interaction	KDS-ST-05-EY-N
6	M2	OK	0.323	steel-interaction	KDS-ST-05-EX-P
7	M5	OK	0.318	steel-interaction	KDS-ST-05-EY-P
8	M8	OK	0.289	steel-interaction	KDS-ST-05-EY-P
9	M12	OK	0.288	steel-interaction	KDS-ST-05-EX-P
10	M4	OK	0.287	steel-interaction	KDS-ST-05-EY-N

11	M11	OK	0.285	steel-interaction	KDS-ST-05-EX-N
12	M1	OK	0.277	steel-interaction	KDS-ST-05-EY-N
13	M18	OK	0.263	steel-interaction	KDS-ST-05-EY-P
14	M22	OK	0.260	steel-interaction	KDS-ST-05-EY-P
15	M35	OK	0.258	steel-deflection	KDS-ST-05-EX-N
16	M36	OK	0.257	steel-deflection	KDS-ST-05-EX-P
17	M21	OK	0.253	steel-interaction	KDS-ST-05-EY-P
18	M19	OK	0.248	steel-interaction	KDS-ST-05-EX-P
19	M17	OK	0.248	steel-interaction	KDS-ST-05-EY-N
20	M33	OK	0.246	steel-deflection	KDS-ST-05-EY-N

Member Design Trace Matrix

Item	Value
Checked members	84
Unimplemented members	0
NG / WARN	0/0
Max utilization	0.363

Member	Type	Status	Util.	Formula	Action
M6	steel	OK	0.363	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M7	steel	OK	0.329	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M3	steel	OK	0.328	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M10	steel	OK	0.326	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M9	steel	OK	0.323	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M2	steel	OK	0.323	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M5	steel	OK	0.318	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M8	steel	OK	0.289	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M12	steel	OK	0.288	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M4	steel	OK	0.287	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

M11	steel	OK	0.285	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M1	steel	OK	0.277	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M18	steel	OK	0.263	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M22	steel	OK	0.260	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M35	steel	OK	0.258	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M36	steel	OK	0.257	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M21	steel	OK	0.253	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M19	steel	OK	0.248	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M17	steel	OK	0.248	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M33	steel	OK	0.246	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	84	0	0.363
Connections	84	24	2.686
Foundations	12	24	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.